Experiment-7

AIM:

To write a Program to implement flow Control at data Link Layer using SLIDING WINDOW Protocol. Simulate the flow of frames from one node to another

Code:

sender.py

```
import time
w = int(input("Enter window size: "))
msg = (input("Enter message:")
frames = [[i, ch] for i, ch in enumerate(msg)]
base = 0
while base < len(frames):
    with open("sender_Buffer.txt","w") as f
    for i in range(base, min(base+w, len(frames)
        f.write(f"{frames[i][0]}
                {frames[i][1]}\n")
    Print("\n In frames Sent:")
    for i in range(base, (min(base+w, len(frames))
        Print(frames[i])
    time.sleep(3)
try:
    with open("Reciver-Buffer.txt","r") as f:
        data = f.read().strip()
    if data.startswith("Ack"):
        ack = int(data.split(":")[1])
    if ack == base+w-1 or ack >= len(frames)-1
        Print("ACK Recived", ack)
        base = ack+1
    else:
        Print("Wrong ACK Resending...")
```

```
elif data.startswith("NACK"):
        Print("NACK Recived. Resending.....")

empt:
    Print("waiting for Reciver ....")
```

Reciver.Py

```
exp=0
with open("sender_Buffer.txt","r") as f:
        Lines = f.readLines()

    ok = True
    for line in lines:
        no,data = (line.strip()).split(",")
        if int(no)!=exp:
            OK=false:
            have =exp
            break
        empt=1

with open("Reciver_Buffer.txt","w") as f:
    if ok:
        f.write(f"ACK:{exp-1}")
        Print("ACK sent:",exp-1)

    else:
        f.write(f"NACK:{have}")
        Print("NACK Sent:",have)
```

Input:

Enter window size: 3

Enter message : HELLO

Output (sender)

Frames sent:

[0, 'H']

[1, 'E']

[2, 'L']

Ack received : 2.

Frames sent:

[3, 'L']

[4, 'O']

Ack recived : 4

Output (Reciver)

Ack sent : 2

Ack sent : 4

Result:

Thus the program to implement flow control at data link layer using sliding window has been executed successfully.